Technology Stack

Date	10 June 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	2 Marks

Define Your Technology Stack:

The technology stack includes **React.js** for the frontend, **FastAPI (Python)** for the backend, **SQLite** with **SQLAlchemy ORM** for database management, **Google Gemini API** for AI-powered financial analysis and negotiation support, and **Git/GitHub** for version control. This combination enables the development of an intelligent, secure, and scalable **AI Powered Debt Relief & Financial Recovery Platform**.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	**React.js, HTML5, CSS3, JavaScript**	Builds a responsive and user-friendly interface for borrowers to manage loans, monitor financial health, and access AI-powered features.
2	**Backend / Server-Side**	**FastAPI (Python)**	Handles REST APIs, business logic, loan processing, financial analysis, and communication between the frontend, database, and AI services.
3	**Database / Data Storage**	**SQLite, SQLAlchemy ORM**	Stores borrower information, loan

			details, negotiation history, financial reports, and settlement records securely while SQLAlchemy simplifies database operations.
4	**AI & Machine Learning**	**Google Gemini API (Generative AI)**	Generates personalized debt settlement recommendations, professional negotiation letters, financial insights, and borrower guidance using Generative AI.
5	**Cloud / Hosting / Deployment**	**Google Cloud Platform (GCP) / Local Deployment**	Hosts the application, enables scalable deployment, and securely integrates AI services through cloud infrastructure.
6	**Version Control & Collaboration**	**Git, GitHub**	Manages source code, version history, collaboration among team members, and deployment workflows.
7	**Development Tools**	**Visual Studio Code, Postman**	VS Code is used for coding and debugging, while Postman is used for testing REST APIs.
8	**Third-Party APIs / Libraries**	**Google Gemini API, SQLAlchemy, Pydantic**	Provides AI capabilities, ORM support for database management, and data validation for FastAPI applications.